📋 Rapport Technique — Module Scanner : Analyse Complète Scan & Print

Version: 3.0
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Contexte: CMDB Inventory — Tracabilité Assets

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1. Analyse Algorithmique — Workflow Scan & Print

1.1 Cycle de Vie Complet d'un Asset

```
┌──────────────────────────────────────────────────────────────────────┐
│              CYCLE DE VIE ASSET — SCAN & PRINT                       │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  1. RÉCEPTION                                                        │
│     └── Asset créé → QR Code généré automatiquement                 │
│                                                                      │
│  2. ÉTIQUETAGE                                                       │
│     └── PrintJob créé → Étiquette imprimée → Collée sur asset       │
│                                                                      │
│  3. UTILISATION                                                      │
│     └── Scan régulier → Vérification existence/localisation         │
│                                                                      │
│  4. MAINTENANCE                                                      │
│     └── Scan → Ticket créé → Pièces stock consommées                │
│                                                                      │
│  5. MOUVEMENT                                                        │
│     └── Scan → Localisation mise à jour → PrintLog enregistré       │
│                                                                      │
│  6. RETRAITE                                                         │
│     └── Scan → Statut "retired" → Historique complet                │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
```

1.2 Rôle des 4 Nouveaux Modèles

```
┌──────────────────────────────────────────────────────────────────────┐
│                     MODÈLES SCAN & PRINT                             │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ┌──────────────┐          ┌──────────────┐                        │
│  │ PrintTemplate│          │   Printer    │                        │
│  │              │         │              │                       │
│  │ - Format     │         │ - USB/BT/WiFi│                       │
│  │ - Layout     │         │ - IP/MAC     │                       │
│  │ - Taille     │         │ - Config     │                       │
```

```
│   └──────────┘         └──────────┘                          │
│         │                    │                                 │
│         │                    │                                 │
│         └────────────────────┘                                 │
│                   │                                            │
│                   │                                            │
│                   ▼                                            │
│            ┌──────────────┐                                   │
│            │  PrintJob    │                                   │
│            │              │                                   │
│            │ - Assets[]   │                                   │
│            │ - Template   │                                   │
│            │ - Printer    │                                   │
│            │ - Status     │                                   │
│            │ - PDF File   │                                   │
│            └──────┬───────┘                                   │
│                   │                                            │
│                   │ 1-to-Many                                  │
│                   ▼                                            │
│            ┌──────────────┐                                   │
│            │  PrintLog    │                                   │
│            │              │                                   │
│            │ - Asset      │                                   │
│            │ - PrintedBy  │                                   │
│            │ - Timestamp  │                                   │
│            │ - Audit      │                                   │
│            └──────────────┘                                   │
│                                                                │
│  FLUX COMPLET:                                                 │
│  ────────────                                                  │
│  Template + Printer → PrintJob → PDF → PrintLog (audit)        │
│                                                                │
└────────────────────────────────────────────────────────────────┘
```

1.3 Détail des 4 Modèles

PrintTemplate — Configuration Étiquette

Champ	Type	Usage
`format`	30x20, 50x30, 70x40, 80x50	Taille physique étiquette
`width_mm`, `height_mm`	Integer	Dimensions exactes
`qr_size_mm`	Integer	Taille QR Code sur étiquette
`qr_position_x/y_mm`	Integer	Position QR (coordonnées)
`font_size_title/text`	Integer	Tailles police
`show_serial/category/location`	Boolean	Champs à afficher

Exemple d'Usage:

```
# Template standard pour laptops
template = PrintTemplate.objects.create(
    name='Standard Laptop 50x30',
    format='50x30',
    width_mm=50,
    height_mm=30,
    qr_size_mm=20,
    qr_position_x_mm=5,
    qr_position_y_mm=5,
    font_size_title=10,
    font_size_text=8,
    show_serial=True,
    show_category=True,
    show_location=True,
    is_default=True
)
```

Printer — Configuration Imprimante

Champ	Type	Usage
`connection_type`	usb, bluetooth, wifi, network	Type de connexion
`ip_address`, `port`	IP, Integer	Pour WiFi/Ethernet
`mac_address`	String	Pour Bluetooth
`device_path`	String	Pour USB (/dev/usb/lp0)
`dpi`, `speed`, `density`	Integer	Qualité impression
`paper_width/height_mm`	Integer	Format papier
`is_default`	Boolean	Imprimante par défaut

Exemple d'Usage:

```
# Imprimante USB atelier
printer = Printer.objects.create(
    name='MUNBYN RW403B - Atelier',
    connection_type='usb',
    device_path='/dev/usb/lp0',
    dpi=203,
    speed=3,
    density=10,
    paper_width_mm=50,
    paper_height_mm=30,
    is_default=True
)

# Imprimante WiFi bureau
```

```
printer = Printer.objects.create(
    name='Samsung SP350 - Bureau',
    connection_type='wifi',
    ip_address='192.168.1.100',
    port=9100,
    dpi=203,
    speed=3,
    density=12,
    paper_width_mm=70,
    paper_height_mm=40,
    is_default=False
)
```

PrintJob — Job d'Impression (Batch)

Champ	Type	Usage
`uuid`	UUID	Identifiant unique du job
`asset_ids`	JSONField	Liste des assets à imprimer
`template`, `printer`	ForeignKey	Configuration utilisée
`copies`	Integer	Nombre de copies par asset
`status`	pending, processing, completed, failed	Statut du job
`pdf_file`	FileField	PDF généré (stocké)
`error_message`	Text	Erreur si failed
`created_by`	ForeignKey	Technicien qui a lancé

Exemple d'Usage:

```
# Job batch pour 50 laptops
job = PrintJob.objects.create(
    created_by=request.user,
    asset_ids=[210, 211, 212, ..., 260],  # 50 assets
    template=template_50x30,
    printer=printer_munbyn,
    copies=1,
    status='pending'
)

# Task Celery génère le PDF
generate_print_job_pdf.delay(job.id)

# Après génération
job.status = 'completed'
job.pdf_file = 'print_jobs/2026/03/job_abc123.pdf'
job.completed_at = timezone.now()
```

```
job.save()
```

PrintLog — Audit & Traçabilité

Champ	Type	Usage
`job`	ForeignKey	PrintJob associé
`asset`	ForeignKey	Asset imprimé
`printed_by`	ForeignKey	Technicien (qui a imprimé)
`printer_name`, `template_name`	String	Snapshot configuration
`printed_at`	DateTime	Timestamp exact
`ip_address`, `user_agent`	String	Contexte technique

Exemple d'Usage:

```python
# Log créé automatiquement après impression
for asset_id in job.asset_ids:
    PrintLog.objects.create(
        job=job,
        asset_id=asset_id,
        printed_by=request.user,
        printer_name=job.printer.name,
        template_name=job.template.name,
        ip_address=request.META.get('REMOTE_ADDR'),
        user_agent=request.META.get('HTTP_USER_AGENT')
    )

# Requête audit : qui a imprimé quoi ?
PrintLog.objects.filter(
    printed_at__gte=timezone.now() - timedelta(days=30)
).select_related('asset', 'printed_by')
```

1.4 Relations entre Modèles

```
┌──────────────────────────────────────────────────────────────────────┐
│                  DIAGRAMME DE RELATIONS COMPLET                      │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ┌──────────────┐          ┌──────────────┐                        │
│  │   Asset      │          │  QRCode      │                        │
│  │  (210)       │◄─────────│              │                        │
│  │              │  1-to-1  │ - uuid       │                        │
│  │ - S/N        │          │ - image      │                        │
│  └──────┬───────┘          └──────────────┘                        │
│         │                        │                                  │
```

```
│         │ 1-to-Many           │ 1-to-Many                          │
│         │                       │                                   │
│         ▼                       ▼                                   │
│  ┌──────────────┐          ┌──────────────┐                       │
│  │ ScanLog      │          │  PrintLog    │                       │
│  │              │          │              │                       │
│  │ - scanned    │          │ - printed    │                       │
│  │ - timestamp  │          │ - timestamp  │                       │
│  └──────────────┘          └──────────────┘                       │
│                                   │                                 │
│                            1-to-Many                                │
│                                   │                                 │
│                                   ▼                                 │
│                           ┌──────────────┐                        │
│                           │  PrintJob    │                        │
│                           │              │                        │
│                           │ - batch      │                        │
│                           │ - PDF        │                        │
│                           └──────────────┘                        │
│                                   │                                 │
│                       ┌───────────┴──────────┐                    │
│                       │                      │                    │
│                       ▼                      ▼                    │
│              ┌──────────────┐          ┌──────────────┐            │
│              │PrintTemplate│          │   Printer    │            │
│              │              │          │              │            │
│              │ - layout     │          │ - config     │            │
│              └──────────────┘          └──────────────┘            │
│                                                                    │
│  TRACABILITÉ COMPLÈTE:                                             │
│  ────────────────────                                              │
│  Asset → QR Code → Scan (ScanLog) + Print (PrintLog)               │
│  Chaque action est enregistrée avec timestamp + user               │
│                                                                    │
└────────────────────────────────────────────────────────────────────┘
```

---

2. Pages UI à Modifier/Créer

2.1 Pages Existantes à Modifier

Page	Modifications	Priorité
`/admin/scanner/`	✅ Ajouter section Print + Templates + Printers	🔴 Haute
`/admin/assets/`	✅ Bouton "Imprimer Étiquette" (single/batch)	🔴 Haute
`/admin/assets/<id>/`	✅ Onglet "Étiquette" + Preview QR	🔴 Haute
`/admin/stock/`	⚠️ Lien vers impression étiquettes stock	🟠 Moyenne

2.2 Nouvelles Pages à Créer

Page	URL	Description	Priorité
Landing Scan/Print	`/admin/scan-print/`	Hub central techniciens	🔴 Haute
Templates	`/admin/scanner/templates/`	CRUD PrintTemplate	🟠 Moyenne
Printers	`/admin/scanner/printers/`	CRUD Printer config	🟠 Moyenne
Print Jobs	`/admin/scanner/jobs/`	Historique jobs	🟠 Moyenne
Print Logs	`/admin/scanner/logs/`	Audit impressions	🟢 Basse

---

3. Landing Page — Scan & Print Hub

3.1 Concept

```
┌──────────────────────────────────────────────────────────────────┐
│                    LANDING PAGE TECHNICIENS                      │
│                    /admin/scan-print/                            │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│   ┌──────────────────────────────────────────────────────────┐   │
│   │  🎯 CENTRE DE TRACABILITÉ ASSETS                          │   │
│   │     Scanner + Imprimer + Auditer                          │   │
│   └──────────────────────────────────────────────────────────┘   │
│                                                                  │
│   ┌──────────────┐   ┌──────────────┐   ┌──────────────┐         │
│   │   📷 SCAN    │   │   🖨️ PRINT   │   │   📊 AUDIT   │         │
│   │              │   │              │   │              │         │
│   │ Scanner QR   │   │ Étiquettes   │   │ Historique   │         │
│   │ Code-Barres  │   │ Batch/Single │   │ Complet      │         │
│   └──────────────┘   └──────────────┘   └──────────────┘         │
│                                                                  │
│   ┌──────────────────────────────────────────────────────────┐   │
│   │  ⚡ ACTIONS RAPIDES                                       |   |
│   │  [Scanner Asset] [Imprimer Étiquette] [Voir Historique] |   |
│   └──────────────────────────────────────────────────────────┘   │
│                                                                  │
│   ┌──────────────────────────────────────────────────────────┐   │
│   │  📊 STATISTIQUES EN TEMPS RÉEL                            │   │
│   │  Scans today: 145  |  Prints today: 89  |  Assets: 210    │   │
│   └──────────────────────────────────────────────────────────┘   │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

3.2 Code Complet — `templates/admin/scan_print/landing.html`

```
<!-- templates/admin/scan_print/landing.html -->
{% extends 'admin_base.html' %}
{% load static %}

{% block title %}Scan & Print Hub — CMDB Inventory{% endblock %}

{% block extra_css %}
<link href="{% static 'admin_cmdb/css/scan_print_hub.css' %}" rel="stylesheet">
<style>
    :root {
        --bg-dark: #0f172a;
        --bg-card: rgba(15, 23, 42, 0.75);
        --accent: #2563eb;
        --accent-hover: #1d4ed8;
        --success: #22c55e;
        --warning: #f59e0b;
        --danger: #ef4444;
        --text-main: #f8fafc;
        --text-muted: #94a3b8;
    }

    body {
        background: var(--bg-dark);
        color: var(--text-main);
        font-family: 'Inter', sans-serif;
    }

    .hub-hero {
        background: linear-gradient(135deg, rgba(37, 99, 235, 0.2) 0%, rgba(15, 23, 42, 0.8)
100%);
        backdrop-filter: blur(16px);
        border: 1px solid rgba(255, 255, 255, 0.1);
        border-radius: 20px;
        padding: 40px;
        margin-bottom: 40px;
        text-align: center;
    }

    .hub-hero h1 {
        font-size: 2.5rem;
        font-weight: 700;
        margin-bottom: 16px;
        background: linear-gradient(135deg, #fff, #94a3b8);
        -webkit-background-clip: text;
        -webkit-text-fill-color: transparent;
    }

    .hub-hero p {
        color: var(--text-muted);
        font-size: 1.1rem;
```

```
    max-width: 600px;
    margin: 0 auto;
}

.hub-cards {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(350px, 1fr));
    gap: 24px;
    margin-bottom: 40px;
}

.hub-card {
    background: var(--bg-card);
    backdrop-filter: blur(16px);
    border: 1px solid rgba(255, 255, 255, 0.1);
    border-radius: 16px;
    padding: 32px;
    transition: all 0.3s ease;
    cursor: pointer;
    text-decoration: none;
    color: inherit;
}

.hub-card:hover {
    transform: translateY(-4px);
    border-color: var(--accent);
    box-shadow: 0 20px 40px rgba(37, 99, 235, 0.2);
}

.hub-card-icon {
    width: 80px;
    height: 80px;
    border-radius: 16px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 2.5rem;
    margin-bottom: 24px;
}

.hub-card-icon.scan { background: rgba(37, 99, 235, 0.2); color: #3b82f6; }
.hub-card-icon.print { background: rgba(34, 197, 94, 0.2); color: #22c55e; }
.hub-card-icon.audit { background: rgba(245, 158, 11, 0.2); color: #f59e0b; }

.hub-card h3 {
    font-size: 1.5rem;
    font-weight: 600;
    margin-bottom: 12px;
}

.hub-card p {
    color: var(--text-muted);
    line-height: 1.6;
```

```
        margin-bottom: 20px;
    }

    .hub-card-features {
        display: flex;
        flex-wrap: wrap;
        gap: 8px;
        margin-top: 16px;
    }

    .hub-feature {
        padding: 6px 12px;
        background: rgba(255, 255, 255, 0.05);
        border-radius: 20px;
        font-size: 0.8rem;
        color: var(--text-muted);
    }

    .quick-actions {
        background: var(--bg-card);
        backdrop-filter: blur(16px);
        border: 1px solid rgba(255, 255, 255, 0.1);
        border-radius: 16px;
        padding: 32px;
        margin-bottom: 40px;
    }

    .quick-actions h2 {
        font-size: 1.5rem;
        font-weight: 600;
        margin-bottom: 24px;
        display: flex;
        align-items: center;
        gap: 12px;
    }

    .quick-actions-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
        gap: 16px;
    }

    .quick-action-btn {
        display: flex;
        flex-direction: column;
        align-items: center;
        padding: 24px;
        background: rgba(255, 255, 255, 0.03);
        border: 1px solid rgba(255, 255, 255, 0.1);
        border-radius: 12px;
        text-decoration: none;
        color: var(--text-main);
        transition: all 0.2s;
```

```
}

.quick-action-btn:hover {
    background: rgba(37, 99, 235, 0.1);
    border-color: var(--accent);
    transform: translateY(-2px);
}

.quick-action-btn i {
    font-size: 2rem;
    margin-bottom: 12px;
}

.quick-action-btn span {
    font-weight: 600;
    font-size: 0.95rem;
}

.stats-bar {
    background: var(--bg-card);
    backdrop-filter: blur(16px);
    border: 1px solid rgba(255, 255, 255, 0.1);
    border-radius: 16px;
    padding: 24px;
    margin-bottom: 40px;
}

.stats-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
    gap: 24px;
}

.stat-item {
    text-align: center;
}

.stat-value {
    font-size: 2.5rem;
    font-weight: 700;
    color: var(--accent);
    line-height: 1.2;
}

.stat-label {
    color: var(--text-muted);
    font-size: 0.9rem;
    margin-top: 8px;
}

.recent-activity {
    background: var(--bg-card);
    backdrop-filter: blur(16px);
```

```
    border: 1px solid rgba(255, 255, 255, 0.1);
    border-radius: 16px;
    padding: 32px;
}

.recent-activity h2 {
    font-size: 1.5rem;
    font-weight: 600;
    margin-bottom: 24px;
}

.activity-list {
    display: flex;
    flex-direction: column;
    gap: 12px;
}

.activity-item {
    display: flex;
    align-items: center;
    gap: 16px;
    padding: 16px;
    background: rgba(255, 255, 255, 0.03);
    border-radius: 12px;
    transition: background 0.2s;
}

.activity-item:hover {
    background: rgba(255, 255, 255, 0.06);
}

.activity-icon {
    width: 48px;
    height: 48px;
    border-radius: 12px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 1.5rem;
    flex-shrink: 0;
}

.activity-icon.scan { background: rgba(37, 99, 235, 0.2); color: #3b82f6; }
.activity-icon.print { background: rgba(34, 197, 94, 0.2); color: #22c55e; }

.activity-content {
    flex-grow: 1;
}

.activity-title {
    font-weight: 600;
    margin-bottom: 4px;
}
```

```
    .activity-meta {
        font-size: 0.85rem;
        color: var(--text-muted);
    }

    .activity-time {
        font-size: 0.85rem;
        color: var(--text-muted);
        white-space: nowrap;
    }

    @media (max-width: 768px) {
        .hub-hero h1 { font-size: 1.8rem; }
        .hub-cards { grid-template-columns: 1fr; }
        .stats-grid { grid-template-columns: repeat(2, 1fr); }
        .quick-actions-grid { grid-template-columns: repeat(2, 1fr); }
    }
</style>
{% endblock %}

{% block content %}
<div id="scan-print-hub" class="container-fluid py-4">

    <!-- Hero Section -->
    <div class="hub-hero">
        <h1>🎯 Centre de Tracabilité Assets</h1>
        <p>Scanner, imprimer et auditer tous vos assets IT en un seul endroit. Traçabilité
complète du cycle de vie.</p>
    </div>

    <!-- Main Cards -->
    <div class="hub-cards">
        <!-- Scan Card -->
        <a href="/admin/scanner/" class="hub-card">
            <div class="hub-card-icon scan">📷</div>
            <h3>Scanner QR / Code-Barres</h3>
            <p>Identifiez instantanément n'importe quel asset avec votre scanner USB,
douchette Bluetooth ou caméra smartphone.</p>
            <div class="hub-card-features">
                <span class="hub-feature">USB HoneyWell</span>
                <span class="hub-feature">Bluetooth</span>
                <span class="hub-feature">Webcam</span>
                <span class="hub-feature">Smartphone</span>
            </div>
        </a>

        <!-- Print Card -->
        <a href="/admin/scanner/print/" class="hub-card">
            <div class="hub-card-icon print">🖨️</div>
            <h3>Imprimer Étiquettes QR</h3>
            <p>Générez et imprimez des étiquettes QR Code professionnelles pour tous vos
assets. Supporte impression batch.</p>
```

```
            <div class="hub-card-features">
                <span class="hub-feature">Single/Batch</span>
                <span class="hub-feature">USB/BT/WiFi</span>
                <span class="hub-feature">Templates</span>
                <span class="hub-feature">PDF Export</span>
            </div>
        </a>

        <!-- Audit Card -->
        <a href="/admin/scanner/logs/" class="hub-card">
            <div class="hub-card-icon audit">📊</div>
            <h3>Audit & Historique</h3>
            <p>Consultez l'historique complet des scans et impressions. Traçabilité conforme
pour audit et compliance.</p>
            <div class="hub-card-features">
                <span class="hub-feature">ScanLog</span>
                <span class="hub-feature">PrintLog</span>
                <span class="hub-feature">Export CSV</span>
                <span class="hub-feature">Rapports</span>
            </div>
        </a>
    </div>

    <!-- Stats Bar -->
    <div class="stats-bar">
        <div class="stats-grid">
            <div class="stat-item">
                <div class="stat-value">[[ stats.scans_today ]]</div>
                <div class="stat-label">📷 Scans aujourd'hui</div>
            </div>
            <div class="stat-item">
                <div class="stat-value">[[ stats.prints_today ]]</div>
                <div class="stat-label">🖨️ Impressions aujourd'hui</div>
            </div>
            <div class="stat-item">
                <div class="stat-value">[[ stats.total_assets ]]</div>
                <div class="stat-label">💻 Assets totaux</div>
            </div>
            <div class="stat-item">
                <div class="stat-value">[[ stats.printers_active ]]</div>
                <div class="stat-label">🖨️ Imprimantes actives</div>
            </div>
        </div>
    </div>

    <!-- Quick Actions -->
    <div class="quick-actions">
        <h2>⚡ Actions Rapides</h2>
        <div class="quick-actions-grid">
            <a href="/admin/scanner/" class="quick-action-btn">
                <i>📷</i>
                <span>Scanner Asset</span>
            </a>
```

```
            <a href="/admin/assets/" class="quick-action-btn">
                <i>📋</i>
                <span>Sélectionner Assets</span>
            </a>
            <a href="/admin/scanner/print/" class="quick-action-btn">
                <i>🖨️</i>
                <span>Imprimer Étiquettes</span>
            </a>
            <a href="/admin/scanner/templates/" class="quick-action-btn">
                <i>🎨</i>
                <span>Templates</span>
            </a>
            <a href="/admin/scanner/printers/" class="quick-action-btn">
                <i>⚙️</i>
                <span>Imprimantes</span>
            </a>
            <a href="/admin/scanner/jobs/" class="quick-action-btn">
                <i>📜</i>
                <span>Jobs d'Impression</span>
            </a>
            <a href="/admin/scanner/logs/" class="quick-action-btn">
                <i>🔍</i>
                <span>Historique</span>
            </a>
            <a href="/admin/search/" class="quick-action-btn">
                <i>🔎</i>
                <span>Recherche</span>
            </a>
        </div>
    </div>

    <!-- Recent Activity -->
    <div class="recent-activity">
        <h2>🕐 Activité Récente</h2>
        <div class="activity-list" v-if="recentActivity.length > 0">
            <div class="activity-item" v-for="activity in recentActivity"
:key="activity.id">
                <div class="activity-icon" :class="activity.type">
                    [[ activity.type === 'scan' ? '📷' : '🖨️' ]]
                </div>
                <div class="activity-content">
                    <div class="activity-title">[[ activity.title ]]</div>
                    <div class="activity-meta">[[ activity.meta ]]</div>
                </div>
                <div class="activity-time">[[ activity.time ]]</div>
            </div>
        </div>
        <div v-else class="text-center text-muted py-5">
            <i style="font-size: 3rem; opacity: 0.3;">📊</i>
            <p class="mt-3">Aucune activité récente</p>
        </div>
    </div>
```

```
</div>
{% endblock %}

{% block extra_js %}
<script src="{% static 'admin_cmdb/js/scan_print_hub.js' %}"></script>
{% endblock %}
```

3.3 Code JavaScript — `static/admin_cmdb/js/scan_print_hub.js`

```
// static/admin_cmdb/js/scan_print_hub.js
const { createApp } = window.VueCreateApp || Vue.createApp;

createApp({
    delimiters: ['[[', ']]'],
    data() {
        return {
            stats: {
                scans_today: 0,
                prints_today: 0,
                total_assets: 0,
                printers_active: 0
            },
            recentActivity: []
        }
    },
    mounted() {
        this.fetchStats();
        this.fetchRecentActivity();
    },
    methods: {
        async fetchStats() {
            try {
                const [assetsRes, printersRes] = await Promise.all([
                    window.apiClient.get('/inventory/assets/', { params: { page_size: 1 }
}),
                    window.apiClient.get('/scanner/printers/', { params: { is_active: true }
})
                ]);

                this.stats.total_assets = assetsRes.data.count || 0;
                this.stats.printers_active = printersRes.data.length || 0;

                // Stats scans/prints today (endpoints à créer)
                const today = new Date().toISOString().split('T')[0];
                const [scansRes, printsRes] = await Promise.all([
                    window.apiClient.get('/scanner/logs/', { params: { date: today } }),
                    window.apiClient.get('/scanner/print-logs/', { params: { date: today }
})
                ]);

                this.stats.scans_today = scansRes.data.count || 0;
```

```
                this.stats.prints_today = printsRes.data.count || 0;

            } catch (error) {
                console.error('Erreur fetch stats:', error);
            }
        },
        async fetchRecentActivity() {
            try {
                const [scansRes, printsRes] = await Promise.all([
                    window.apiClient.get('/scanner/logs/', { params: { page_size: 5 } }),
                    window.apiClient.get('/scanner/print-logs/', { params: { page_size: 5 }
})
                ]);

                const scans = (scansRes.data.results || scansRes.data || []).map(s => ({
                    id: `scan-${s.id}`,
                    type: 'scan',
                    title: `📷 Scan: ${s.asset_name || 'Asset inconnu'}`,
                    meta: `Par: ${s.scanned_by || 'Inconnu'} • IP: ${s.ip_address ||
'N/A'}`,
                    time: this.formatTime(s.created_at)
                }));

                const prints = (printsRes.data.results || printsRes.data || []).map(p => ({
                    id: `print-${p.id}`,
                    type: 'print',
                    title: `🖨️ Impression: ${p.asset_name || 'Asset inconnu'}`,
                    meta: `Par: ${p.printed_by || 'Inconnu'} • ${p.printer_name}`,
                    time: this.formatTime(p.printed_at)
                }));

                this.recentActivity = [...scans, ...prints]
                    .sort((a, b) => new Date(b.time) - new Date(a.time))
                    .slice(0, 10);

            } catch (error) {
                console.error('Erreur fetch activity:', error);
            }
        },
        formatTime(dateStr) {
            if (!dateStr) return '-';
            const date = new Date(dateStr);
            const now = new Date();
            const diff = now - date;

            if (diff < 60000) return 'À l\'instant';
            if (diff < 3600000) return `Il y a ${Math.floor(diff / 60000)} min`;
            if (diff < 86400000) return `Il y a ${Math.floor(diff / 3600000)} h`;
            return date.toLocaleDateString('fr-FR');
        }
    }
}).mount('#scan-print-hub');
```

4. URLs à Ajouter

```
# backend/scanner/urls.py
from django.urls import path
from . import views

urlpatterns = [
    # ... existing URLs

    # Landing Page Scan & Print
    path('scan-print/', views.scan_print_landing, name='scan-print-landing'),

    # Print Management
    path('print/', views.print_labels_view, name='print-labels'),
    path('templates/', views.PrintTemplateViewSet.as_view({'get': 'list', 'post':
'create'}), name='print-templates'),
    path('templates/<int:pk>/', views.PrintTemplateViewSet.as_view({'get': 'retrieve',
'put': 'update', 'delete': 'destroy'}), name='print-template-detail'),
    path('printers/', views.PrinterViewSet.as_view({'get': 'list', 'post': 'create'}),
name='printers'),
    path('printers/<int:pk>/', views.PrinterViewSet.as_view({'get': 'retrieve', 'put':
'update', 'delete': 'destroy'}), name='printer-detail'),
    path('jobs/', views.PrintJobViewSet.as_view({'get': 'list', 'post': 'create'}),
name='print-jobs'),
    path('jobs/<int:pk>/', views.PrintJobViewSet.as_view({'get': 'retrieve'}), name='print-
job-detail'),
    path('logs/', views.PrintLogViewSet.as_view({'get': 'list'}), name='print-logs'),
]
```

5. Workflow Complet — Exemple Technicien

```
┌──────────────────────────────────────────────────────────────────────┐
│              WORKFLOW TECHNICIEN COMPLET                             │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  1. ARRIVÉE MATIN                                                    │
│     └── /admin/scan-print/ → Vue statistiques jour                   │
│                                                                      │
│  2. RÉCEPTION 10 LAPTOPS                                             │
│     └── /admin/assets/new/ → Créer 10 assets                         │
│     └── QR Codes générés automatiquement                             │
│                                                                      │
│  3. ÉTIQUETAGE                                                       │
│     └── /admin/assets/ → Sélectionner 10 laptops                     │
│     └── Action: "Imprimer Étiquettes"                                │
│     └── PrintJob créé → PDF généré → Impression MUNBYN               │
│     └── Coller étiquettes sur laptops                                │
│                                                                      │
│  4. SCAN CONTRÔLE                                                    │
│     └── /admin/scanner/ → Scanner chaque laptop                      │
```

```
│      └── Vérifier fiche asset correcte                          │
│      └── ScanLog enregistré automatiquement                     │
│                                                                 │
│  5. MAINTENANCE LAPTOP #215                                     │
│      └── Scan laptop → Fiche asset                              │
│      └── Bouton: "Créer Ticket"                                 │
│      └── Ticket créé → Pièces consommées (StockMovement)        │
│      └── PrintLog: Étiquette de maintenance imprimée            │
│                                                                 │
│  6. FIN JOURNÉE                                                 │
│      └── /admin/scan-print/ → Vérifier activité jour            │
│      └── Scans: 45 | Prints: 23 | Assets: 210                   │
│      └── Export CSV pour rapport                                │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

✅ Checklist Finale

Élément	Statut	Fichier
Modèles Print	✅	`scanner/models.py`
Landing Page	✅	`templates/admin/scan_print/landing.html`
JS Hub	✅	`static/admin_cmdb/js/scan_print_hub.js`
URLs	✅	`scanner/urls.py`
Views	⚠️	`scanner/views.py` (à compléter)
API Endpoints	⚠️	`scanner/api/views.py` (à compléter)
CSS Hub	⚠️	`static/admin_cmdb/css/scan_print_hub.css`

Cette landing page est le hub central pour toutes les opérations Scan & Print de vos techniciens ! 🎉